AMIT SINGH PARIHAR

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SUMMARY

AI/ML Developer with 2 years of experience building LLM-based systems at TCS — primarily around code migration tooling and automated documentation pipelines. Comfortable working with LangGraph, RAG architectures, FastAPI, and open-source models (Qwen, Llama). Most of my work involves turning rough prototypes into something that actually runs reliably in a constrained environment.

TECHNICAL SKILLS

LLM & Orchestration: LangGraph, LangChain, OpenAI API, HuggingFace, Prompt Engineering, tool calling, interrupt/resume flows

RAG & Retrieval: FAISS, ChromaDB, Pinecone, PGVector, cross-encoder re-ranking, semantic search, query rewriting, CRAG

Backend & Deployment: FastAPI, Flask, REST APIs, AWS (Bedrock, SageMaker, S3, EC2, Lambda), Streamlit, Docker (basic)

DS/ML/AI - Generative AI, Statistics, Machine Learning, Deep Learning, NLP, DS lifecycle

Languages & Tools: Python, SQL, Git, Ollama, Scikit-learn, Pandas, NumPy, TensorFlow, Power BI

EXPERIENCE

GenAI Developer | Tata Consultancy Services (TCS) | *Aug 2024 – Present*

Java → Kotlin Migration Tool

- Built a LangGraph pipeline that takes a Java project and outputs Kotlin — covering file scanning, chunking, code generation, dependency rewriting, and compile-time validation. Reduced manual migration effort by ~60%.
- After each generation pass, the output was compiled and linted automatically; failed builds fed the error back into the LLM for up to 3 retries before flagging for human review.
- Ran entirely offline on internal TCS servers using Qwen3-Coder via Ollama — no source code left the environment.
- Reused the same approach for VB6→Python, ASP.NET→ASP.NET Core, Java version upgrades, and React migrations.

Automated BRD/FSD Documentation Pipeline

- Built a RAG pipeline that reads legacy VB6 and Java codebases and generates Business Requirement Documents and Functional Specification Documents. Cut documentation prep time by ~65%.
- Used Qwen3 embeddings in pgvector with cross-encoder re-ranking; this pushed retrieval precision ~25% above plain cosine similarity on internal benchmarks.
- Replaced token-count chunking with logical-unit chunking (classes, modules, function groups) to avoid splitting context across function boundaries.

UI Migration — Vision-Language Integration

- Extended the migration pipeline to handle UI changes: took new design mockups, passed them through a vision-language model (Gemma) to extract layout and component structure, then used that output to drive changes to legacy frontend code.
- Reduced iteration cycles between developers and designers during UI migration sprints.
- Mentored junior team members joining the GenAI team — walked through LangGraph basics, RAG design choices, and pipeline debugging.

Python Developer Intern | ART – Abstinent Research & Technologies | *Jan 2024 – Jul 2024*

- Built a natural language-to-SQL system using few-shot prompting that let non-technical users query a MySQL database in plain English — supported multi-table joins and reduced incorrect query generation significantly.

CERTIFICATIONS

- AWS Certified Cloud Practitioner • Generative AI: From GANs to CLIP (Udemy) • Data Analytics & Visualization — Accenture Virtual Experience • Foundation to AI, Data Science & Data Analysis — Samatrix Consulting

EDUCATION

Bachelor of Computer Applications (BCA) — CGPA: 8.34

Specialization: AI & Data Analytics | LNCT University, Bhopal | Sept 2021 – Jun 2024